INCH-POUND

MIL-DTL-43492D
19 December, 2000
SUPERSEDING
MIL-T-43492C
20 August 1987

DETAILED SPECIFICATION

TENT SECTIONS, FRAME TYPE, EXPANDABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tent sections for an expandable, frame-supported tent of wind-resistant cotton sateen (see 6.1). Minimum dimensions of the expandable tent are as follows: Length 16 feet; width 16 feet; ridge height 8 feet, 6 inches; eave height 6 feet. Also included are a cover for the 16 foot by 16 foot tent, and arch and purlin carrying covers for the tent frame sections.

1.2 Classification. (see 6.2)

Type I - End section
Type II - Intermediate section

2. APPLICABLE DOCUMENTS

*2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CNR, 700 Robbins Ave., Philadelphia, PA 19111-5096 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-S-125	- Screening, Insect, Nonmetallic
V-F-106	- Fasteners, Slide, Interlocking
V-T-285	- Thread, Polyester
TT-P-595	- Preservative Coating, Canvas
DDD-L-20	- Label: For Clothing, Equipage and Tentage (General Use)
A-A-3174	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
A-A-50057	- Rope, Fibrous, Tent Lay
A-A-55126	- Fastener tapes, Hook and Loop, Synthetic

MILITARY

MIL-C-496	- Clips, End Strap
MIL-W-530	- Webbing, Textile, Cotton, General Purpose
MIL-H-639	- Hooks (for Belts, Straps, and Equipage Items)
MIL-L-1709	- Lines, Tent
MIL-S-1734	- Slips, Tent Line
MIL-R-2327	- Rings, Connecting, Round
MIL-W-4088	- Webbing, Textile, Woven, Nylon
MIL-F-10884	- Fasteners, Snap
MIL-L-11075	- Loops, Strap Fastener
MIL-PRF-20696	- Cloth, Coated, Polyester or Nylon, Waterproof
MIL-R-30500	- Rope, Polyester
MIL-PRF-44103	- Cloth, Duck, Polyester,
MIL-DTL-43002	- Shield, Stovepipe, Tent, Nonmetallic
MIL-C-43256	- Cord, Polyester, Solid Braid
MIL-T-43566	- Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-W-44049	- Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or in Colors

STANDARDS

MILITARY

MS 70101	- Adapter, Reversible, Quick Fit, 500 lb.; 1 ¾ Inch

(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the Document Automation and Production Service (DAPS), Bldg. 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF THE ARMY FIELD MANUAL

FM 10-16 – General Repair of Tents, Canvas and Webbing

(Copies of the manual are available from the U. S. Army Biological and Chemical Command, ATTN: SSCNC-EMSS, Natick, MA 01760-5017.)

DRAWINGS

U.S. ARMY BIOLOGICAL CHEMICAL COMMAND

5-4-783	- Tent, General Purpose, Small; Cover
5-4-961	- Tent Sections, Frame Type, Expandable; Assembly
5-4-962	– Intermediate Section
5-4-963	– End Section
5-4-964	– End Section – Sections
5-4-965	– Window Details
5-4-966	– Window Sections
5-4-967	– Purlin and Heater Duct Details and Sections
5-4-968	– Intermediate Vent Assembly
5-4-969	– Intermediate Vent Sections
5-4-970	– Stovepipe Opening Details and Sections
5-4-971	– Becket Loop and Guy Line Lug Details, Sections and Enlarged Views
5-4-972	– End Vent Assembly
5-4-973	– End Vent Sections
5-4-974	- Door Detail and Sections
5-4-1043	- Carrying Cover, Arch
5-4-1044	- Carrying Cover, Purlin
5-4-1760	- Erection Instructions

(Copies of drawings are available from the U. S. Army Biological Chemical Command, ATTN: AMSSB-REO-S(N), NATICK, MA 01760-5017.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-2016	- Standard Test Methods for Moisture Content of Wood
ASTM-D-6193	- Standard Practice for Stitches and Seams

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

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AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA

NASM16491 - Grommet, Metallic, General Specification For

(Applications for copies of referenced documents should be addressed to the Aerospace Industries Association of America, Inc., 1250 Eye Street, N. W. Washington, DC 20005-3924.)

*2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.2 Materials and components.

3.2.1 Cloth.

3.2.1.1 Cloth, Duck.. Cloth for the tent sections shall conform to MIL-PRF-44103, Class 1, Grade A & C with the following exception:

a. the maximum weight shall be increased to 16 oz/yd^2.

3.2.1.2 Cloth, coated. Cloth for the reinforcements, tent covers, frame covers, purlin flaps and the sod cloths shall conform to Type II, Class 2 of MIL-PRF-20696. The color shall be camouflage green 483.

3.2.2 Tape, Cotton. Tape shall conform to type I, class 4; ¾, 1, 1-1/2 and 2 inch widths of MIL-T-43566. For covers only, as an alternate, the tape shall be polypropylene conforming to type I, class 4 of MIL-W-44049. The color of the tapes shall be camouflage green 483.

3.2.3 Webbing.

3.2.3.1 Webbing, cotton. Cotton webbing shall conform to type IIb, class 4, 1 inch width of MIL-W-530. The color shall be camouflage green 483.

3.2.3.2 Webbing, nylon. Nylon webbing shall conform to type VIII, class 1 or class 2 of MIL-W-4088. The color shall be camouflage green 483.

3.2.4 Thread, polyester. Thread shall conform to type I, class 1, subclass B of V-T-285. The dyed thread shall show colorfastness to weathering equal to or better than the standard sample (see 6.5). The color shall be camouflage green 483. When no standard shade sample is available, the dyed thread shall show good colorfastness to weathering. Thread size shall be as follows:

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a. For all stitching except zig-zag stitching and stitching indicated by "Z" on drawings size FF for needle and bobbin.
b. For zig-zag stitching, size E for needle and bobbin.
c. For stitching indicated by "Z" on drawing size FF for needle and size F for looper.

Alternate Thread size "E" for needle/bobbin and "F" for needle may be used for bartacking, box stitch formations and "W" stitch formations.

*3.2.5 Fastener tape. Hook and loop fastener tape shall conform to 1 or 2 inch widths, type II hook, class 3 of A-A-55126. The color shall be camouflage green 483.

3.2.6 Lines, tent. Tent lines shall conform to types I, XXI and XXXV of MIL-L-1709 except that the type XXXV lines shall be finished in accordance with figure 3 of MIL-L-1709. Tent lines in types specified may be fabricated from sisal, polyester or duel fiber polyethylene/polyester as alternates. Polyester rope shall be in accordance with MIL-R-30500.

3.2.7 Stiffener, rope. Rope stiffeners for the window flaps and drip caps shall conform to type I or II, class 2, 1 inch circumference of MIL-R-1670.

3.2.8 Cord, polyester. Cord for the becket lines, vent lines, window flap lines and toggle loops shall be 1/8 inch diameter, conforming to MIL-C-43256. The color shall be camouflage green 483.

3.2.9 Screening, Non-Metallic The screening shall be Type III, class 1 or 2, size 18 X 18, color green of L-S-125, or Type II, class 2, size 18 X 18 color green of L-S-125.

*3.2.10 Film, plastic. Plastic film shall conform to the requirements for 0.0060 inch thick film, type I, class 1, grade A, B or C, finish 1 of A-A-3174 except that the film thickness shall be 0.020 $\pm$ 0.004 inch. The requirements for slip, ink adhesion, water vapor permeability, impact resistance and heat seal strength shall not apply and the elongation at break, minimum percent (TD) shall be 290.

*3.2.11 Shield, pipe. Stovepipe shields shall conform to MIL-DTL-43002.

3.2.12 Fasteners, snap. Snap fasteners shall conform to style 2, finish 2, button 24 line size, size 1 of MIL-F-10884.

3.2.13 Fasteners, slide, interlocking. The slide fastener shall conform to size H of V-F-106. The slide fastener chain shall be fabricated of brass or polyester coil. The slider for the brass chain shall be brass and the slider for the polyester coil shall be die case zinc alloy. The length of the slide fastener chain shall be such that after the slide fastener is sewed on, the tape shall not be puckered and the slider shall operate freely. The color of the slide fastener tape shall be camouflage green 483. The tape shall be water repellent and mildew resistant treated. The tape shall show good fastness to weathering. Alternatively, the slide fastener chain shall be polyester continuous monofilament in a ladder or coil type configuration. The color of the chain, slider, pull and stops shall approximate the shade of the tape.

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As an alternate to using preassembled slide fasteners, a continuous chain application may be utilized as part of the manufacturing method. The method requires that the entire chain tape and tape extensions be securely stitched with the required seam without any puckering or waviness. The slider shall be indented into position to provide for a smooth running chain disengagement with the slider mouth squarely against the bottom staple stop and for a smooth running chain engagement with the slider throat evenly against the top stops and with no excess chain element showing. If continuous monofilament polyester chain is used, sewn top stops without gapping are allowed.

3.2.13.1 Flap, window. Window flap fasteners shall be type I, style 18, single wire stirrup pull.

3.2.13.2 Fasteners, door. Door fasteners shall be type III, style 6 of V-F-106, double wire stirrup pull.

3.2.14 Grommet, metallic. Grommets shall conform to type III, class 3, size Nos. 0, 1, 3, 4 and 5 of MIL-G-16491.

3.2.15 Hooks, "S". "S" hooks shall conform to type XII, class 2, size G of MIL-H-639.

*3.2.16 Slips, tent line. Tent line slips shall conform to type II of MIL-S-1734. As an alternate, "S" hooks conforming to Chicago Hardware and Fixture Style No. S-2-A (Zinc Plated, .148 wire size, 1 ½ inches length) or equal may be substituted.

3.2.17 Loops, strap fastener. Strap loops shall conform to type III style B of MIL-L-11075.

3.2.18 Clips, end. End clips shall conform to type I, class I or 2, 1 inch of MIL-C-496.

3.2.19 Adapter. Adapters for the carrying covers shall conform to the material, design and dimensions specified on MS 70101.

3.2.20 Rings, connecting. Connecting rings shall conform to class 2 or 5, size 4 of MIL-R-2327.

3.2.21 Toggles, wood. Wood toggles shall be of the design and dimensions shown on Drawing 5-4-971. The toggles shall be fabricated from straight-grained, sound and smooth lumber and shall contain not more than 15% moisture when tested as specified in 4.4.1.1. The toggles shall be free from knots, shakes, checks, splits, splintery surfaces, or case hardening. Any of the following woods may be used:

Ash	Hickory	Elm, rock	Pecan
Beech	Maple, hard	Birch, yellow	Oak, red or white

3.3 Construction. Construction shall conform to the drawings listed in section 2 and as specified herein.

3.3.1 Stitching, machine.

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*3.3.1.1 Types. All stitch types, except bartacking, shall conform to ASTM-D-6193 as follows:

a. For all stitching except zig-zag stitching and stitching indicated by "Z" on drawings and for attachment of slide fasteners – type 301, 5 to 7 stitches per inch.

b. For zig-zag stitching - type 304, 10 to 14 stitches per inch.

c. For stitching indicated by "Z" on drawings and for attachment of slide fasteners – type 401, 5 to 7 stitches per inch. Chain portion of stitching shall not appear on outside of tent or covers.

d. For bartacking – 5/8 ± 1/16 inch long, 1/8 inch minimum bight and containing 28 stitches.

3.3.1.2 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1 inch at each break on stitch type 301, not less than ¾ inch at each break on stitch type 304 and not less than 1 ½ inches at each break on stitch type 401. Thread breaks in type 401 stitching may be overstitched with stitch type 301. Bartacks shall be free of thread breaks.

3.3.1.3 Stitching ends. The ends of 301 stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem or held down by other stitching. Where 301 stitching is performed automatically on stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking, not less than three tying, overlapping , or back stitches shall be used to secure the ends of stitching. The ends of type 304 stitching shall be firmly secured by backstitching or overstitching or by increasing the number of stitches per inch at the ends for a minimum of ¼ inch.

3.3.1.4 Skipped stitches. Two or more consecutively skipped stitches occurring in type 301 or 304 stitching shall be overstitched not less than 1 inch. Any skipped stitches in type 401 stitching shall be overstitched not less than 1 ½ inches. Skipped stitches in 401 stitching may be overstitched with type 301 stitching.

3.3.1.5 Thread lubricant. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited.

3.3.2 Splicing of cloth. There shall be no splicing of tent end, intermediate sections or cover panels. Splicing of reinforcements and sod cloth sections will be allowed provided no piece is less than 24 inches long. Pieces shall be joined as shown on Drawing 5-4-962.

3.3.3 Attachment of wood toggles. Attachment of wood toggles shall be as illustrated on drawings. Wrap cord around the toggle 1 ½ times and secure with an overhand knot in a manner that prevents removal of toggle.

3.3.4 Attachment of slide fasteners. Care shall be exercised to trim all thread ends to ½ inch or less and to maintain specified distances between edges of fabric attached to slide fastener tape so that there will be no interference with the operation of the slider. The length of the slide fastener chain shall be such that after the slide fastener is sewn on, the tape shall not be puckered and the slider shall operate freely.

3.3.5 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. Grommets shall be securely clinched without cutting the materials. Roll of the clinched grommet shall not contain more than three scores (splits).

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3.3.6 Setting of end clips. Webbing shall be inserted the full depth of the clip. Clips shall be securely attached without cutting the webbing and the webbing shall lie flat at the point of entry.

3.3.7 Wicking of sewing thread (cup test). There shall be no dripping of water through the stitching when the tent section is tested as specified in 4.5.2. Seepage or leakage of water at a few points caused by hydrostatic pressure through needle holes or between plies of a lap seam shall not be scored as a failure. The seam leakage test is intended to insure that nonwicking thread has been utilized and that no lubrication has been added to the polyester thread prior to or during the sewing operation.

*3.3.8 Cementing. Cementing shall be in accordance with the following procedure:

a. Surface preparation. Surfaces to be bonded shall be clean and dry prior to application of the adhesives. Coated cloth surfaces to be bonded shall be thoroughly cleaned by rubbing with toluene-wetted clean cloths or pads, which may contain a mild, bonding abrasive of 80, or finer, grit size or may be of a course weave (as for scouring). If abrading is exercised, there shall be no scuffing or abrading of the base cloth. Separate clean cloths and clean solvent shall be used for final cleaning wipes. Abrasive coated cloths or pads shall not be used directly on stitching at seams.

b. Adhesive mixing. The ratio of the adhesive mixture shall be as specified on the drawings. Prior to extracting quantities of the base and accelerator to be mixed, any settlement shall be uniformly redispersed. Containers for mixing, transfer, and use shall be clean prior to each filling. Adhesive in the containers shall be covered during transit to work stations and during sustained periods of non-use to prevent loss of solvent by evaporation. After mixing, the adhesive shall be used only while sufficiently spreadable to obtain an even coat.

c. Adhesive application. A minimum of two coats of mixed adhesive shall be evenly spread on the cleaned surfaces to be bonded. The first coat shall be dry to the touch before the application of the second coat. Prior to assembly, the adhesive coated surfaces shall not be allowed to be contaminated by a visible layer of dust or other foreign matter. Applied adhesive shall be free of heavy smears or excessively thick lap edges (resulting from insufficient spreading), thin starved areas, uncoated areas or other defects characteristic of improper application. The second coat shall be dried as appropriate for the assembly method to be used; i.e., to a tacky (not wet) aggressive grab condition for the two-way dry stick method; or to a low-tack (dry condition) for two-way stick, solvent reactivation method.

d. Assembly. Assembly of components shall be made when the tack and grab of the adhesive coated surfaces are in a suitable range. Either excess solvent in the adhesive layers or excessive drying or open time shall be avoided. Surfaces shall be joined in the locations specified on the drawings, applying adequate pressure and rolling or barring firmly to produce an adequate bond. Care shall be exercised to avoid finger contact in the adhesive coated areas along the edges of seams, patches or other bonded parts. Precautions should be taken to prevent adhesive application, assembling, or curing of an assembly at a temperature below 60° F. The assembled work shall not be moved to soon or in such a way as to result in edge lifting or decreasing strength of bond until the adhesive has set. Surfaces shall be assembled on the same day that they were coated, or a third coat of adhesive shall be applied to each surface. When surfaces have been assembled in an incorrect location, the adhesive bond may be immediately separated and reset at the proper location. To facilitate separation, toluene may be applied at the bond line by squirt can or atomizer and the component then peeled away as the adhesive softens. The excess toluene solvent shall be allowed to evaporate before resetting the surfaces. When incorrectly located parts are separated and the adhesive is not in an active condition or has transferred to the other surface, the affected surface shall be given an additional coat of adhesive before reassembling. Any adhesive residue shall not extend further than ½ inch beyond the edges of seams strapping, splices, patches or other bonded parts.

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*3.3.9 Repairs to polyester fabric. All necessary repairs shall be effected in accordance with Department of the Army Field Manual FM 10-16, except that hand-sewn patches or cemented patches other than as specified herein will not be acceptable. Cemented patches on holes, cuts or tears not more than 1 inch in diameter or length, using cement per Minnesota Mining EC-847 or equal (see 6.11) shall be permitted. A maximum of three cemented patches per item will be allowable without prior contracting officer authorization. Each patch shall be large enough to overlap the damaged area with a margin of at least ¾ inch on all sides. Cemented patches shall be placed on the outside of the tent and shall not be permitted over seams or in hardware areas. All other repairs shall require authorization by the contracting officer. Repairs required as a result of failure of the cup test and when authorized by the contracting officer shall be performed by treating with a compound conforming to type I, class 2 of TT-P-595 prior to retesting (see 6.6).

*3.3.10 Repairs to vinyl coated fabric. Repairs to holes, cuts, or tears and repairs of burn-throughs resulting from the heat sealing operation, when necessary, shall be effected as follows:

a. No more than one repair to holes, cuts or tears shall be allowed per cover or floor section.

b. No more than two repairs to burn-throughs shall be allowed per cover or floor section.

c. Repairs shall be accomplished by the application of heat sealed or cemented patches.

d. For repairs of burn-throughs on panel lap seams, the width of the patch shall be such that the overlap on each side of the burn-through shall be equal to the width of seam. The maximum length of the patch shall not exceed the length of the sealing die plus the width of seam allowance at each end. All patches for repairs other than burn-throughs shall not exceed the size specified herein.

e. Patches shall be applied to the top side of the cover or floor section and the patched area shall be smooth, i.e., with the torn or cut edges securely and completely sealed or cemented.

3.4 Marking. All markings and labels shall conform to DDD-L-20.

3.4.1 Special marking. The letters “US” or the item nomenclature on the end sections, intermediate sections, frame carrying covers, and tent cover shall conform to type IV, class 9 in the size characters and locations shown on the drawings. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.4.2 Identification labels. Identification labels shall conform to type VI, Class 6 and shall be 3 ½ inches by 5 ½ inches minimum with a blank space of at least 1 ¼ inches provided on the bottom of the label for the Department of Defense Inspector’s stamp. Identification labels shall be stitched to the end sections, intermediate sections, frame carrying covers and tent cover in the locations shown on the drawings. Identification information may also be stenciled directly on the frame carrying covers and shall conform to type IV, class 6, ½ inch high in the locations shown on Drawings 5-4-1043 and 5-4-1044.

3.4.3 Erection instruction labels. Labels of erection instructions shall conform to type VI, class 7 and shall be stitched to the tent and tent cover in the locations shown on the drawings. The label shall be a proportionate copy of Drawing 5-4-1760 reproduced from a glossy print of the drawing furnished by the contracting officer. The size label shall be, borderline to borderline, 15 inches $\pm$ ¼ inch in the long dimension, with a minimum of ½ inch margin outside of the borderline on all sides.

3.4.4 Workmanship. Cloth components shall be clean and free of holes, cuts, tears, or cloth defects such as multiple floats, broken or missing yarns, or areas untreated with compound. Webbing

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and tapes shall not have frayed or scalloped edges and rope components shall be free of cuts or breaks. Thread tension shall be maintained so that there will be no loose stitching, and seam allowances shall be maintained with seams properly sewn so that no run-offs, twists, pleats or open seams shall result. Care shall be taken in sewing to see that no needle chews occur. All thread ends shall be trimmed to ½ inch or less. Metal components shall be free of burrs, sharp edges and corroded areas and shall not be broken or malformed.

*3.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in 4.4.3 and 4.4.4 and tested as specified in 4.4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.3 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.4. Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

*4.4.1 Testing of components. In addition to the quality assurance provisions of the subsidiary documents and drawings, the test specified in 4.5.1 shall be performed on wood toggles specified in 3.2.21 for determination of moisture content. The lot shall be expressed in units of toggles. The sample unit shall be one wood toggle. There shall be one determination per sample unit and the result shall be reported to the nearest 0.1 percent. The inspection level shall be S-1.

4.4.2 In-process inspection. Inspection shall be made to determine that the following are as specified:

a. There shall be no lubrication of polyester thread prior to or during sewing operations for compliance with 3.3.1.5,

b. Holes punched to receive grommets are in compliance with 3.3.5.

Whenever nonconformance is noted, correction shall be made to the items affected and the process.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table I. The lot size shall be expressed in units of completely fabricated end sections, intermediate sections, tent covers or frame carrying covers, as applicable. The sample unit shall be one end section, intermediate section, tent cover or frame carrying cover, as applicable. The inspection level shall be I.

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TABLE I. Defects

Examine	Defect	Classification Major	Minor A	Minor B
Cloth	Cut, tear, or hole (including exposed drill hole) greater than 1/8 inch in length or diameter:			
	- on tent section	101		
	- on cover or sod cloth		201	
	Cut, tear or hole (including exposed drill hole) 1/8 inch or less in length or diameter:			
	- more than 20 holes:			
	- on tent section	102		
	- on cover or sod cloth		202	
	- more than 10 but not more than 20 holes:			
	- on tent section		203	
	- on cover or sod cloth			301
	- more than 5 but not more than 10 holes:			
	- on tent section			302
	Untreated area on either side of cloth more than 1 inch in length or diameter		204	
	Broken or missing yarn; multiple floats clearly visible at normal inspection distance, (approx 3 feet)		205	
Screening	Cut or tear:			
	- more than 2 inches in length	103		
	- ¼ inch or less in length			303
	- more than ¼ inch in length but not more than 2 inches in length		206	
Webbing, tie tapes	Edge frayed or scalloped			304
	Cut or tear			305
Tape omitted:				
	- one			306
	- two or more		207	
Hardware	Broken	104		
	Malformed:			
	- fails to perform intended function	105		
	- but will perform intended function		208	
	Corroded area			307
	Burr or sharp edge that may cause injury in handling or damage to fabric	106		
	Not specified type or size		209	
Grommets	Improperly set, two or more teeth exposed (except where a grommet is set as specified but is in an area of uneven thickness which prevents uniform clinching)			308
	Clinched loosely, allowing grommet to rotate around hole			309
	Clinched excessively tight (i.e., cutting cloth)	107		

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TABLE I. Defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	Insecurely clinched (i.e., grommets or washers become disengaged)	108		
	Set with barrel inserted from outside instead of inside:			
	- on eave or ridge		210	
	- at any other location			310
	Roll of a grommet contains more than three scores (splits)			311
End clips	Improperly clinched (i.e., webbing not flat at point of entrance into clip)			312
	Clinched excessively tight, cutting webbing or loosely clinched (i.e., clip becomes disengaged from webbing)		211	
Wood toggles	Contains knot, shakes, split, or splintery surface		212	
Rope components tent lines	Any cut or break:			
	- on ridge guy line		213	
	- on eave guy line		214	
	- on any other line			313
	Eye splice omitted		215	
	Eye splice not as specified			314
	Free end not finished as specified			315
Cord	Not stitched as specified	109		
Window opening	Not reinforced all around with 2-inch facing		216	
Window pane	Any puncture, deep crease, or crack	110		
	Pane not fully caught in facing stitching all around		217	
Seams & stitching	Open seam on stitch type 301:			
	- for more than ¼ inch but not more than 1inch			316
	- for more than 1 inch		218	
	Open seam on stitch type 401:			
	- one row of stitching open for 1 inch or less			317
	- one row of stitching open for more than 1 inch:			
	- on intermediate section		219	
	- on end section or cover			318

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TABLE I. Defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	Both rows of stitching open (not necessarily at the same location):			
	- on intermediate section	111		
	- on end section or cover		220	
	Note: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or run-off stitches occur. On double stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Raw edges caught in stitching:			
	- more than 1 inch but not more than 3 inches			319
	- more than 3 inches		221	
	Thread breaks:			
	- overstitched less than 1 ½ inches on stitch type 401			320
	- overstitched less than 1 inch on stitch type 301			321
	- overstitched less than ¾ inch on stitch type 304			322
	Note: Thread breaks not overstitched shall be classified as open seams.			
	Needle chews resulting in cut, tear, or hole (see cloth defects for cut, tear or hole)			
	Skipped stitches:			
	- overstitched less than 1 ½ inches on stitch type 401			323
	- overstitched less than 1 inch on stitch type 301 or 304 (when 2 or more skipped stitches occur consecutively)			324
	Note: Skipped stitches not overstitched shall be classified as open seams.			
	One or more required rows of stitching omitted (except on boxstitching)	112		
Boxstitching	Incomplete:			
	- one row of stitching omitted			325
	- two or more rows of stitching omitted		222	

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TABLE I. Defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	Less than specified length by more than 1/8 inch			326
Bartacking	Less than specified length by more than 1/8 inch			327
	Width less than 1/8 inch		223	
Stitching ends	Overstitched less than 1 inch on stitch type 301 (except where ends are held down by other stitching, turned under in a hem, or where stitching is performed automatically)		224	
	Secured with less than three tying, overlapping or back stitches when automatic stitching is performed			328
	Secured with less than ¼ inch of backstitching, overstitching or an increased number of stitches per inch at the end of stitch type 304			329
Seams	Pleated:			
	- on tent section		225	
	- on cover			330
Seam type	Wrong seam type	113		
Stitch type	Wrong stitch type	114		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches		226	
	Tight, as evidenced by puckering on fabric, for more than 6 inches		227	
Stitches per inch	Less than the minimum specified (except on panel joining seams):			
	- one stitch			331
	- two or more stitches		228	
	Less than the minimum specified on panel joining seams:		229	
	More than the maximum specified:			
	- one stitch			332
	- two or more stitches		230	
Note:	Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners shall be classified as follows:			
	a. With the minor "B" defect classification: - no defect			

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TABLE I. Defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	b. With the minor "A" defect classification: - minor "B" defect			
Stitching gage	Not as specified		231	
Stitching margin	Larger than specified by more than 3/16 inch: - for more than 2 inches but not more than 4 inches in length			333
	- for more than 4 inches in length		232	
	Less than specified: - for more than 2 inches but not more than 4 inches in length			334
	- for more than 4 inches in length		233	
Stitching indicated by "Z"on drawings	Chain portion of stitching shows on outside of tent section or cover	115		
Toggle stitching	Loose tension	116		
	Not secured as specified, for example toggle can be removed	117		
	Number of stitches: - more than specified			335
	- less than specified	118		
Mends, darns or patches	Repairs not in accordance with Department of the Army Field Manual FM 10-16 or not authorized by the contracting officer (see 3.3.9)	119		
Components & assembly	Any component or part omitted, not assembled or fabricated of applicable reference material	120		
Hems	Constructed with partial turnunder when double turnunder is required		234	
	Twisted or pleated			336
Reinforce-ments	Improperly applied, causing excessive fullness on reinforcement or reinforced parts	121		
Cloth Splicing	Splice in tent section or cover panel	122		
	Spliced pieces not joined as shown on Drawing 5-4-962			337
Slide fastener	Slider jams or fails to interlock chain	123		
	Any tape improperly assembled, causing failure of chain to effect a smooth and			

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TABLE I. Defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	secure closure	124		
	Thong omitted			338
Stovepipe shield	Set to roof with oval opening in a horizontal instead of vertical position	125		
	Exposed stitching or needle holes			339
Cementing, general	Puckers, loose edges, unadhered pocket or tunnels		235	
	Excess adhesion resulting in sags or lumps evident through outer layer of fabric		236	
	Adhesive exposed beyond the overcoated area on interior surface			340
General	Any other construction detail not performed as specified		237	
Marking & labels	Omitted, incorrect, illegible, or misplaced			341
Cleanness	Grease or oil stains			342
	Thread ends not trimmed to ½ inch or less (except in slide fastener area)			343
	Thread ends not trimmed to ½ inch or less in slide fastener area		238	

4.4.4 End item dimensional examination. The end item shall be examined for the defects listed in Table II. The lot size shall be expressed in units of completely fabricated end sections, intermediate sections, tent covers or frame carrying covers, as applicable. The sample unit shall be one end section, intermediate section, tent cover or frame carrying cover, as applicable. The inspection level shall be S-3.

TABLE II. End item dimensional defects

Examine	Defect	Classification Major	Minor A	Minor B
Overall dimensions of end sections & frame carrying covers	Smaller than specified dimensions less applicable minus tolerances indicated on drawings	101		
	Larger than specified dimensions and applicable plus tolerances indicated on drawings			301
Covers, tent	Smaller than specified dimensions less applicable minus tolerance indicated on drawing but not smaller than the specified dimensions less			

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TABLE II. End item dimensional defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	twice the applicable minus tolerance			302
	Smaller than specified dimensions less twice the applicable minus tolerances indicated on drawing		201	
	Larger than specified dimensions and applicable plus tolerances indicated on drawings			303
Location dimensions (unless otherwise specified)	Not within specified tolerance: - not exceeding twice the applicable tolerance			304
	- exceeding twice the applicable tolerance		202	
Webbing, tie tapes	Finished length less than specified by more than 1 inch			305
Grommets	Set beyond specified location by more than ½ inch			306
Rope components; tent lines	Eye on footstop or line less than specified by more than ½ inch		203	
	Finished length of guy line less than specified: - by not more than 4 inches			307
	- by more than 4 inches but not more than 12 inches		204	
	- by more than 12 inches	102		
Beckets	Misplaced: - by more than ¼ inch but not more than ½ inch			308
	- by more than ½ inch		205	
	Length of loop more or less than specified by ½ inch or more		206	
Hems	Hems required to finish less than ¾ inch in width, less than specified; hems required to finish more than ¾ inch in width, less than specified by more than ¼ inch	103		
Reinforcements	Finished dimension less than specified by more than ½ inch			309
	Any spliced piece less than 24 inches in length			310

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TABLE II. End item dimensional defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
Sod cloths	Less than specified width by more than one inch			311
	Any spliced piece less than 24 inches in length			312
Window opening	Vertical center line out of alignment with center line of blanket:			
	- by more than ½ inch but not more than one inch			313
	- by more than 1 inch	104		
	Two inch facing uneven with edge of opening by more than ¼ inch		207	
Ventilator opening	Misplaced: - by more than ½ inch			314
Stovepipe opening	Misplaced: - by more than ½ inch			315
Tent cover flaps	Center line of flaps set off center of body by more than ¾ inch			316

4.4.5 End item testing. The completely fabricated tent section shall be tested as specified in 4.5.2 for compliance with 3.3.7. The lot shall be expressed in units of completely fabricated tent sections. The sample unit shall be one completely fabricated end section or intermediate section, as applicable. The inspection level shall be S-3. Each sample unit shall be tested in two specific areas and the results shall be reported as pass or fail. Failure in any area shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for marking, materials, workmanship and content defects.

4.5 Methods of inspection.

4.5.1 Moisture content of wood toggles. The moisture content of wood toggles shall be determined in accordance with the oven-drying method referenced in ASTM D 2016.

4.5.2 Wicking of sewing thread (cup test). The tent sections shall be tested in two areas as follows: At a roof panel lap seam and on an eave purlin or header reinforcement seam.

4.5.2.1 Procedure. Suspend the seamed section of the test fabric in the center of a 6 $\pm$ 1/8 inch diameter hoop and form a depression with the seamed fabric in the hoop to a depth of 1 $\pm$ ¼ inch. Slowly pour 500 ml of water at 77° $\pm$ 4°F into the cupped area (depression) and observe the under surface of the fabric for water penetration. Any wicking of water along the sewing thread, identified by a discoloration or darkening of the thread within 5 minutes after water is poured shall constitute a failure. Water leakage through the needle holes or between the plies of the lapped seam fabric shall not be considered a failure.

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5. PACKAGING

*5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

*(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tent sections are designed to be used in forming a 16 foot by 16 foot frame supported tent for use as a division tactical operations and planning center, communications center and briefing tent. The tent is easily expandable in 16 foot increments.

*6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.2 and 6.4).
d. Levels of preservation and packing (see 5.1).

6.3 Sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Components. The following components are required for the erection of the 16 foot by 16 foot tent:

Component	Quantity	National Stock No.
Frame section, tent end section	2	8340-00-234-4708
Frame section, tent intermediate section	1	8340-00-234-4712
Pin, tent, aluminum, 9 inch	18	8340-00-261-9749
Pin, tent, steel, 12 inch	6	8340-00-823-7451
Pin, tent, wood, 16 inch	10	8340-00-261-9750

*6.6 Repair. A satisfactory method of applying the compound for the repair of leaking seams is as follows:

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Observe the following warnings and apply K Kote Seam Sealer per P/N 83-243C or equal (see 6.11) to fabric assemblies as described below.

WARNING

Do not smoke or use seam sealer or solvent near open flame.

Be sure to observe all warnings and instructions that come with sealer and solvent.

Use seam sealer and solvent in an open, well ventilated area, away from sources of combustion. Avoid direct sunlight if possible.

NOTE

Tent can be erected or fabric assemblies can be removed and laid flat.

Do not use a foam brush. Use a bristle brush only.

Sealer may be applied to outside or inside of fabric assemblies.

Sealer should only be applied to new stitches, patches, etc., or those areas that have shown water leakage.

Two light coats of sealant are more effective than one thick coat. Do not use excessive amounts of sealer, as this does not increase effectiveness and will give the tent a poor appearance.

Do not leave original sealer container open, as sealer will thicken and become unusable. Limit the amount of time container is open.

(1) Surface to be sealed should be cleaned as thoroughly as practical and completely dried before applying sealer.
(2) Pour small amount of sealer into a clean container and immediately reseal the gallon container.
(3) Apply sealer to stitching or repair to be sealed, overlapping seam stitching or edges of repair by ½ inch.
(4) Allow first coat to dry to the touch.
(5) Apply second coat as described in step (3).

6.6.1 Seams to be treated. Seams that have failed the test specified in 4.5.2 should be treated as follows:

a. Failure of panel lap seam – treat all panel lap seams.

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b. Failure of purlin or header reinforcement stitching – treat all stitching on reinforcements.

6.7 International standardization agreements. Certain provisions of this document are the subject of international standardization agreements as cited in NATO, STANAG NO. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revision or cancellation of this document is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.8 Subject term (key word) listing.
Shelter, expandable
Sections

*6.9 Supply source. Alternate hooks described in 3.2.16 may be obtained from the Chicago Hardware and Fixture Co., 9100 Park Avenue, Franklin Park, IL 60131.

*6.10 Seam sealer supply source. A source of supply for seam sealer used in 6.6 is:
Kenyon Consumer Products
P.O. Box 3715
1425 Kingstown Road
Peace Dale, Rhode Island 02883

*6.11 Equal item. Prior to the use of an "or equal" item, the supplier must submit the item, with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.12 Changes from previous issue. The margins of this document have been marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Air Force – 99

Review activities:
Army – MD
Air Force – 11.6, 82

Preparing activity
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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-43492D	**2. DOCUMENT DATE** ***(YYYYMMDD)*** 2000/SEP/19

3. DOCUMENT TITLE
TENT SECTIONS, FRAME TYPE, EXPANDABLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-CNRP	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN (215) 737-3290 444-3290
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave. (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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